

Name : M.Koushik Reddy

Reg.no: 192372185

CSA1509-Cloud computing and Big Data Analytics

EXPERIMENT -- 23

Attach Virtual Block to a Virtual Machine

Aim

To attach a virtual block (virtual hard disk) to a virtual machine and verify that the data stored in it remains intact even after the virtual machine is released or shut down.

Algorithm

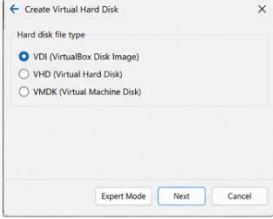
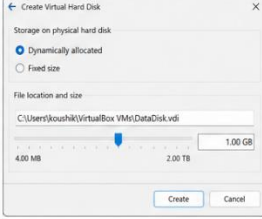
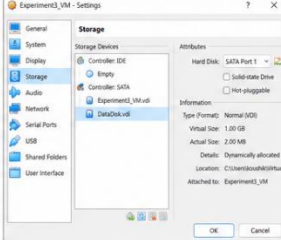
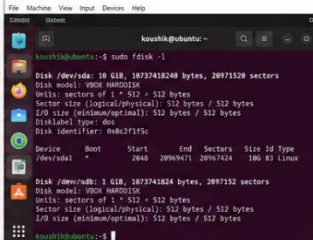
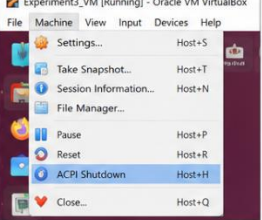
1. Open **Oracle VM VirtualBox**.
2. Create a new virtual hard disk (VDI) with the required storage size.
3. Select the virtual machine and open **Settings** → **Storage**.
4. Attach the newly created virtual disk to the VM.
5. Start the virtual machine and initialize/mount the attached disk.
6. Create a file and write some data into the virtual disk.
7. Unmount the disk and shut down the virtual machine.
8. Start the virtual machine again and mount the same virtual disk.
9. Open the previously created file and verify the stored data.
10. Compare the data before and after releasing the VM to confirm data persistence.

Result

The virtual block was successfully attached to the virtual machine. The data stored in the virtual disk remained intact after shutting down and restarting the virtual machine. Hence, the virtual block **retains data even after the release of the virtual machine**.

Step by Step :-

23. Write a Procedure To Attach Virtual Block To The Virtual Machine And Check Whether It Holds the Data Even After The Release Of The Virtual Machine

1. Create a Virtual Disk (Virtual Block) Go to File → Tools → Media Manager → Hard disks Click "Add" to create a new virtual disk. 	2. Configure Virtual Disk Choose "Dynamically allocated" and set size (e.g., 1 GB). Click "Create". 	3. Attach Virtual Disk to VM Select the VM → Settings → Storage → Controller: SATA → Click "Add Hard Disk" → Choose the created virtual disk. 	4. Start the Virtual Machine Start the VM and initialize the attached disk inside the OS (Ubuntu). 
5. Create File and Write Data Create a file on the new disk and write some data. <pre>koushik@ubuntu:~\$ sudo mkfs.ext4 /dev/sdb mkfs: 1.45.5 (8P-Jun-2023) Creating filesystem with 262144 4k blocks Filesystem UUID: 3d6e9da-dc57-da12-bcfd-1741d4c9e1a1 Superblock backups stored on blocks: 32768, 98304, 163840, 229376 Allocating group tables: done Writing inode tables: done Creating journal (8192 blocks): done Writing superblocks and filesystem accounting information: done koushik@ubuntu:~\$ sudo mkdir /mnt/data koushik@ubuntu:~\$ sudo mount /dev/sdb /mnt/data koushik@ubuntu:~\$ echo "Hello Virtual Disk" sudo tee /mnt/data/test.txt Hello Virtual Disk koushik@ubuntu:~\$ sudo umount /mnt/data</pre>	6. Shut Down and Release the VM Shutdown the VM completely. 	7. Restart the VM Start the VM again and mount the disk to check the data. <pre>koushik@ubuntu:~\$ sudo mount /dev/sdb /mnt/data koushik@ubuntu:~\$ ls /mnt/data test.txt koushik@ubuntu:~\$ cat /mnt/data/test.txt Hello Virtual Disk koushik@ubuntu:~\$</pre>	8. Verify Data Persistence The data is intact even after shutting down and releasing the VM. 

Result: The virtual block (DataDisk.vdi) was attached to the virtual machine, data was written to it, and after shutting down and restarting the virtual machine, the data was found intact. Hence, the virtual block holds the data even after the release of the virtual machine.